

SL 4.5

Let consider a pair of qubits in a composition entangled state $|\Psi\rangle$ and assume that there exist variable (hidden variable) that determine the spin-observables of the qubits in any direction.

$\Rightarrow$ we will derive the BELL inequality and then in a more general form called by CHSH. However, we will prove that there exist entangled state which violates the BELL inequality.

4.5.1 Consider a pair of qubits in the entangled Bell state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|0,1\rangle - |1,0\rangle) \in \mathcal{H}^A \otimes \mathcal{H}^B$ of which A is sent to Alice and B is sent to Bob.

$$\hat{n} = \hat{n}(\theta, \phi) = \begin{pmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{pmatrix} \in \mathbb{R}^3.$$

On her qubit A, she measures $\hat{n}^A \cdot \sigma$

$$\sum_A \hat{n}^A \cdot \sigma = \hat{n}^A \cdot \sigma$$

$$\sum_B \hat{n}^B \cdot \sigma = \hat{n}^B \cdot \sigma \quad (\text{wrt Bob}).$$

Note that... (Exercise 4.54, 4.55)

① Let $|\uparrow_{\hat{n}}\rangle$ and $|\downarrow_{\hat{n}}\rangle$ be defined as in (2.125) and (2.126)

$$\Rightarrow |\Psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow_{\hat{n}}\rangle \otimes |\downarrow_{\hat{n}}\rangle - |\downarrow_{\hat{n}}\rangle \otimes |\uparrow_{\hat{n}}\rangle)$$

$$\textcircled{2} \quad \langle \sum_{\hat{n}^A}^A \otimes \sum_{\hat{n}^B}^B \rangle_{\Psi} = -\hat{n}^A \cdot \hat{n}^B$$

Thm 4.5 Let $S_{\hat{n}}^A$ and $S_{\hat{n}}^B$ be two discrete random variables on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ that are parametrized by unit vectors $\hat{n} \in \mathbb{R}^3$ and takes values in $\{\pm 1\} \Leftrightarrow$

$$S^X: S_{\mathbb{R}^3} \times \Omega \longrightarrow \{\pm 1\} \quad \text{for } X \in \{A, B\}$$

$$(\hat{n}, \omega) \longmapsto S_{\hat{n}}^X(\omega)$$

and which, in addition satisfy for all $\hat{n} \in S_{\mathbb{R}^3}$,

$$\underline{E[S_{\hat{n}}^A S_{\hat{n}}^B]} = -1.$$

$\Rightarrow$ for arbitrary unit vectors $\hat{n}^i$ with $i \in \{1, 2, 3\}$

the BELL inequality holds:

$$\underline{|E[S_{\hat{n}^1}^A S_{\hat{n}^2}^B] - E[S_{\hat{n}^1}^A S_{\hat{n}^3}^B]| - E[S_{\hat{n}^2}^A S_{\hat{n}^3}^B]} \leq 1$$



pt) observe that $-1 = E[S_{\hat{n}}^A S_{\hat{n}}^B]$

$$\begin{aligned} &= P\{S_{\hat{n}}^A = S_{\hat{n}}^B\} - P\{S_{\hat{n}}^A = -S_{\hat{n}}^B\} = 1 - 2P\{S_{\hat{n}}^A = -S_{\hat{n}}^B\} \\ &(\because P\{S_{\hat{n}}^A = S_{\hat{n}}^B\} = 1 - P\{S_{\hat{n}}^A = -S_{\hat{n}}^B\}) \text{ and} \\ &\text{we have } P\{S_{\hat{n}}^A = -S_{\hat{n}}^B\} = 1 \text{ for arbitrary } \hat{n}. \end{aligned}$$

Moreover,

$$\begin{aligned} E[S_{\hat{n}_1}^A S_{\hat{n}_2}^B] - E[S_{\hat{n}_1}^A S_{\hat{n}_3}^B] &= -E[S_{\hat{n}_1}^A S_{\hat{n}_2}^A] + E[S_{\hat{n}_1}^A S_{\hat{n}_3}^A] \\ &= E[S_{\hat{n}_1}^A (S_{\hat{n}_3}^A - S_{\hat{n}_2}^A)] = E[S_{\hat{n}_1}^A ((S_{\hat{n}_2}^A)^2 S_{\hat{n}_3}^A - S_{\hat{n}_2}^A)] \\ &= E[S_{\hat{n}_1}^A S_{\hat{n}_2}^B (S_{\hat{n}_2}^A S_{\hat{n}_3}^A - 1)] \text{ and thus} \\ |E[S_{\hat{n}_1}^A S_{\hat{n}_2}^B] - E[S_{\hat{n}_1}^A S_{\hat{n}_3}^B]| &= |E[S_{\hat{n}_1}^A S_{\hat{n}_2}^A (S_{\hat{n}_2}^A S_{\hat{n}_3}^A - 1)]| \\ &\leq E[|S_{\hat{n}_1}^A S_{\hat{n}_2}^A (S_{\hat{n}_2}^A S_{\hat{n}_3}^A - 1)|] = E[|1 - S_{\hat{n}_2}^A S_{\hat{n}_3}^A|] \\ &= E[1 - S_{\hat{n}_2}^A S_{\hat{n}_3}^B] = 1 + E[S_{\hat{n}_2}^A S_{\hat{n}_3}^B], \end{aligned}$$

where we used $|S_{\hat{n}_1}^A S_{\hat{n}_2}^A| = 1$ and $S_{\hat{n}_2}^A S_{\hat{n}_3}^A \leq 1$.

Remark Let $\hat{n}^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\hat{n}^2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, $\hat{n}^3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ □

and recall $\langle \sum_{\hat{n}^A}^A \otimes \sum_{\hat{n}^B}^B \rangle_{\mathbb{F}^+} = -\hat{n}^A \cdot \hat{n}^B$

then, $|\langle \sum_{a^1}^A \otimes \sum_{a^2}^B \rangle_{\Psi^-} - \langle \sum_{a^1}^A \otimes \sum_{a^3}^B \rangle_{\Psi^-}|$
 $\rightarrow \langle \sum_{a^2}^A \otimes \sum_{a^3}^B \rangle_{\Psi^-} = |\vec{a}^1 \cdot \vec{a}^3 - \vec{a}^1 \cdot \vec{a}^2| + \vec{a}^2 \cdot \vec{a}^3$
 $= |-\frac{1}{\sqrt{2}}| + \frac{1}{\sqrt{2}} = \sqrt{2} > 1$

this is contradiction to the above BELL neg $\star$.

Q. $\star$ or $\star \star$??

4.5.2 CHSH (Generalization of the BELL neg)

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- But CHSH inequality need not the priori condition

$$E[S_a^A S_a^B] = -1.$$

Lemma 4.6 Let S_i with $i \in \{1, 2, 3, 4\}$ be four discrete random variables on a $(\Omega, \mathcal{A}, P)$ that can take only the values $\{\pm 1\}$, that is

$$S_i: \Omega \longrightarrow \{\pm 1\} \text{ for } i \in \{1, 2, 3, 4\}$$

$$w \longmapsto S_i(w)$$

then, $|E[S_1 S_2] - E[S_1 S_3] + E[S_2 S_4] + E[S_3 S_4]| \leq 2$.

Pf) since, we have for all $w \in \Omega$ and $i \in \{1, 2, 3, 4\}$
 $s_i(w) \in \{\pm 1\}$. it follows that.. either

$$\begin{cases} s_2(w) - s_3(w) = 0 \Rightarrow s_2(w) + s_3(w) = \pm 2 & \text{or} \\ s_2(w) + s_3(w) = 0 \Rightarrow s_2(w) - s_3(w) = \pm 2 \end{cases}$$

By using again $s_1(w), s_4(w) \in \{\pm 1\}$, we get

$$s_1(w)(s_2(w) - s_3(w)) + s_4(w)(s_2(w) + s_3(w)) = \pm 2$$

$$\Rightarrow |s_1(w)(s_2(w) - s_3(w)) + s_4(w)(s_2(w) + s_3(w))| \leq 2$$

$$\Downarrow$$

$$E\left(\begin{matrix} \text{ } \\ \text{ } \end{matrix} \right) \leq 2.$$



Now, we consider pairs of particles of which one is accessible to Alice, the other to Bob.

More generally, Alice can select the measurement device by selecting a parameter p^A from a set P of device parameters. This means Alice measures an observable denoted by $S_{p^A}^A$ the measure values by $S_{p^A}^A$ which can take only values $\{\pm 1\}$.
 (Bob) $(S_{p^B}^B)$

Thm 4.7 (CHSH) Let S_p^A and S_p^B be two discrete r.v on a (L.A.P) that can only values $\widehat{\in} \{\pm 1\}$ and that are parametrized by a parameter p in a parameter set P , that is.

$$S^X : P \times \Omega \longrightarrow \{\pm 1\} \quad \text{for } X \in \{A, B\}$$

$$(p, \omega) \longmapsto S_p^X(\omega)$$

$\Rightarrow$ For arbitrary parameters $p_1, \dots, p_4 \in P$ the following CHSH inequality holds:

$$\frac{|E[S_{p_1}^A S_{p_2}^B] - E[S_{p_1}^A S_{p_3}^B] + E[S_{p_4}^A S_{p_2}^B] + E[S_{p_4}^A S_{p_3}^B]|}{\underbrace{\quad \quad \quad}_{\leq 2}} \leq 2$$

pt) If we apply Lemma 4.6 by setting

$$S_i = S_{p_i}^A \quad \text{for } i \in \{1, 4\}, \quad S_i = S_{p_i}^B \quad \text{for } i \in \{2, 3\}.$$

Remark Assume consider two particles that are put in an entangled BELL state $|\Psi^-\rangle$ and as observables $S_{p_i}^X$, the spin-observables $\sum_{\hat{n}^A}^A = \hat{n}^A \cdot \sigma$, $\sum_{\hat{n}^B}^B = \hat{n}^B \cdot \sigma$. In this case $p_i = \hat{n}^i$ and $S_{p_i}^X = \sum_{\hat{n}^i}^X$ for $X \in \{A, B\}$ and $i \in \{1, 2, 3, 4\}$. □

Let $\hat{n}^i = \begin{pmatrix} \cos V_i \\ 0 \\ \sin V_i \end{pmatrix} \in S_{\mathbb{P}^3}^1$ for $i \in \{1, 2, 3, 4\}$

and recall $\langle \Sigma_{\hat{n}^A}^A \otimes \Sigma_{\hat{n}^B}^B \rangle_{\Psi^-} = -\hat{n}^A \cdot \hat{n}^B$.

$$\begin{aligned} \text{then, } \langle \Sigma_{\hat{n}^1}^A \otimes \Sigma_{\hat{n}^2}^B \rangle_{\Psi^-} &= \langle \Sigma_{\hat{n}^1}^A \otimes \Sigma_{\hat{n}^2}^B \rangle \\ &= \langle \Sigma_{\hat{n}^1}^A \otimes \Sigma_{\hat{n}^3}^B \rangle_{\Psi^-} + \langle \Sigma_{\hat{n}^1}^A \otimes \Sigma_{\hat{n}^2}^B \rangle_{\Psi^-} + \langle \Sigma_{\hat{n}^4}^A \otimes \Sigma_{\hat{n}^3}^B \rangle_{\Psi^-} \\ &= -(\cos(V_1 - V_2) + \cos(V_1 - V_3) - \cos(V_4 - V_2) + \cos(V_4 - V_3)) \end{aligned}$$

~~⊙ ⊙ ⊙ ⊙~~

$$V_1 = \frac{3}{4}\pi, \quad V_2 = \frac{\pi}{2}, \quad V_3 = 0, \quad V_4 = \frac{\pi}{4}$$

$$\therefore \textcircled{A}' = -2\sqrt{2} \quad \therefore |-2\sqrt{2}| > 2,$$

this contradicts to ~~⊙ ⊙ ⊙ ⊙~~

Q. ~~⊙ ⊙ ⊙ ⊙~~ or ~~⊙ ⊙ ⊙ ⊙~~ ??

A. Nature behaves in accordance with ~~⊙ ⊙ ⊙ ⊙~~

and violates CHSH inequality. But...

For example, in the state $|\Psi^-\rangle$, we choose to measure spins in the direction $\hat{n}^2 = \hat{n}^1$, then the quantum mechanical prediction ~~⊙ ⊙ ⊙ ⊙~~ gives $-\sqrt{2}$, satisfies CHSH req.

prop 4.8 In any separable state $|Q\rangle \otimes |\Psi\rangle \in \mathcal{H}^A \otimes \mathcal{H}^B$, the expectation values of spin observables $\sum_{a=1}^A \otimes \sum_{b=1}^B$ in arbitrary spin-directions $\hat{n}^i$ with $i \in \{1, 2, 3, 4\}$ satisfy the CHSH variant of the BELL inequality. $\Leftrightarrow |\langle \mathcal{A} \rangle| \leq 2$ holds

H) Generally, the expectation values of products of $M^A \otimes M^B$ factorize in separable states $|Q\rangle \otimes |\Psi\rangle$, we have

$$\begin{aligned} \langle M^A \otimes M^B \rangle_{Q \otimes \Psi} &= \langle Q \otimes \Psi | M^A \otimes M^B (Q \otimes \Psi) \rangle \\ &= \langle Q \otimes \Psi | M^A Q \otimes M^B \Psi \rangle = \langle Q | M^A Q \rangle \langle \Psi | M^B \Psi \rangle \\ &= \langle M^A \rangle_Q \langle M^B \rangle_\Psi \quad \text{--- (1) From (2.118)} \end{aligned}$$

$$|Q\rangle = e^{i\varphi} \cos B |0\rangle + e^{i\tau} \sin B |1\rangle$$

$$|\uparrow_{\hat{n}}(\theta, \varphi)\rangle = e^{-\frac{i\varphi}{2}} \cos \frac{\theta}{2} |0\rangle + e^{\frac{i\varphi}{2}} \sin \frac{\theta}{2} |1\rangle,$$

$$\therefore |Q\rangle = e^{i \cdot \frac{\varphi + \tau}{2}} |\uparrow_{\hat{n}}(2B, \frac{\tau - \varphi}{2})\rangle = e^{\frac{i(\varphi + \tau)}{2}} |\uparrow_{\hat{n}} Q\rangle$$

$$\text{by using } \hat{n}^Q = \hat{n}(2B, \frac{\tau - \varphi}{2}). \text{ Also}$$

$$\therefore |\Psi\rangle = e^{i\delta} |\uparrow_{\hat{n}} \Psi\rangle \text{ with suitably chosen } \delta \text{ and } \hat{n}^\Psi.$$

Exercise 4.57: $\langle \Sigma_n \rangle_{|T_m\rangle} = \hat{n} \cdot \hat{n}$

$\therefore$ ① + Exercise + Irrelevant

$$\Rightarrow \langle \Sigma_{\hat{n}^1}^A \otimes \Sigma_{\hat{n}^2}^B \rangle_{\mathbb{Q} \otimes \Psi} = \langle \Sigma_{\hat{n}^1}^A \rangle_{\mathbb{Q}} \langle \Sigma_{\hat{n}^2}^B \rangle_{\Psi}$$

$$= (\hat{n}^1 \cdot \hat{n}^{\mathbb{Q}}) (\hat{n}^2 \cdot \hat{n}^{\Psi}) \text{ and thus}$$

$$|\langle \otimes' \rangle| = |\hat{n}^1 \cdot \hat{n}^{\mathbb{Q}} (\hat{n}^2 \cdot \hat{n}^{\Psi} - \hat{n}^3 \cdot \hat{n}^{\Psi}) + \hat{n}^4 \cdot \hat{n}^{\mathbb{Q}} (\hat{n}^2 \cdot \hat{n}^{\Psi} + \hat{n}^3 \cdot \hat{n}^{\Psi})|$$

$$\leq |\hat{n}^1 \cdot \hat{n}^{\mathbb{Q}}| |\hat{n}^2 \cdot \hat{n}^{\Psi} - \hat{n}^3 \cdot \hat{n}^{\Psi}| + |\hat{n}^4 \cdot \hat{n}^{\mathbb{Q}}| |\hat{n}^2 \cdot \hat{n}^{\Psi} + \hat{n}^3 \cdot \hat{n}^{\Psi}|$$

$$\leq |(\hat{n}^2 - \hat{n}^3) \cdot \hat{n}^{\Psi}| + |(\hat{n}^2 + \hat{n}^3) \cdot \hat{n}^{\Psi}|$$

$$\leq 2 \cdot \max \{ |\hat{n}^2 \cdot \hat{n}^{\Psi}|, |\hat{n}^3 \cdot \hat{n}^{\Psi}| \} \leq 2$$

$$\left(\because |x| + |y| = \begin{cases} |x+y| & \text{if } xy \geq 0 \\ |x-y| & \text{if } xy \leq 0 \end{cases} \right)$$

□

Sec 4.6 BELL telephone.

* Super-luminal (빛보다 빠른) device can not be used to transmit information at all.

Assume Alice : H^A
 Bob : H^B and

Alices : M^A and $\tilde{M}^A$ (distinct observables)

Also, $\lambda_a \in \sigma(\mu^A)$ and $\tilde{\lambda}_a \in \sigma(\tilde{\mu}^A)$ are non-degenerate in H^A .

$\Rightarrow \exists$ ONB $\{|e_a\rangle\}$ of H^A consisting of $|e_a\rangle \in \text{Eig}(\mu^A, \lambda_a)$
 $\exists // \{|\tilde{e}_a\rangle\} // // \{|\tilde{e}_a\rangle \in \text{Eig}(\tilde{\mu}^A, \tilde{\lambda}_a)$

From Exercise 2.15,

$$|\tilde{e}_a\rangle = U|e_a\rangle = \sum_{a'} \langle e_{a'}|U|e_a\rangle |e_{a'}\rangle = \sum_{a'} U_{a'a} |e_{a'}\rangle$$

($U \in \mathcal{U}(H^A)$)

For Bob's system, let $\{|f_b\rangle\} \in H^B$ are ONB in H^B . Note that the set $\{|e_a \otimes f_b\rangle\}$ and $\{|\tilde{e}_a \otimes f_b\rangle\}$ are ONB in $H^A \otimes H^B$, respectively.

Also, $\{|e_a \otimes f_b\rangle\}$ are eigenvectors of $\mu^A \otimes 1^B$ and $\{|\tilde{e}_a \otimes f_b\rangle\} // // \tilde{\mu}^A \otimes 1^B. \Rightarrow$

we have,

$$\begin{aligned} (\mu^A \otimes 1^B) |e_a \otimes f_b\rangle &= \lambda_a |e_a \otimes f_b\rangle \\ (\tilde{\mu}^A \otimes 1^B) |\tilde{e}_a \otimes f_b\rangle &= \tilde{\lambda}_a |\tilde{e}_a \otimes f_b\rangle \end{aligned}$$

s.t $\sigma(\mu^A \otimes 1^B) = \sigma(\mu^A)$ and $\sigma(\tilde{\mu}^A \otimes 1^B) = \sigma(\tilde{\mu}^A)$

(observe that $\dim \text{Eig}(\mu^A \otimes 1^B, \lambda_a) = \dim H^B$.)

$$|e_a \otimes e\rangle = \sum_b Q_b |e_a \otimes f_b\rangle \quad (\text{eigensate of } M^A \otimes I^B)$$

Similar statement holds for $\tilde{M}^A \otimes I^B$.

Let the composition system be the pure state

$$|\Phi\rangle = \sum_{a,b} \Phi_{ab} |e_a \otimes f_b\rangle = \sum_{a,b} \tilde{\Phi}_{ab} |\tilde{e}_a \otimes f_b\rangle \text{ and}$$

the particles of M^A to distributed to Alice
 M^B // // // Bob.

She measures the observable M^A and this is a measurement of $M^A \otimes I^B$.

If λ_a is the observed value of that measurement

$$\text{then (2.31)} \text{ gives } P_{\lambda_a} = \frac{P_{\lambda_a} \rho_{\Phi} P_{\lambda_a}}{\text{tr}(P_{\lambda_a} \rho_{\Phi})} \quad \begin{matrix} \text{(projection)} \\ \text{(postulate)} \end{matrix}$$

where $P_{\lambda_a} = |e_a\rangle\langle e_a| \otimes I^B$ is projector on to the

Eig($M^A \otimes I^B, \lambda_a$) and $\rho_{\Phi} = |\Phi\rangle\langle\Phi|$ is the density operator.

The mixed state : reduced density operator.

Exercise 4.16 $\rho^B(e) = \sum_{a,b} \sum_a \Phi_{ab} \overline{\Phi_{ab}} |f_b\rangle\langle f_b|$

In order to send the bit 1 to Bob, Alice measures the $\tilde{M}^A$.

$$\rho^B(\tilde{\rho}) = \sum_{i,j} \sum_a \tilde{\Phi}_{a,i} \overline{\tilde{\Phi}_{a,j}} |t_i\rangle\langle t_j|$$

$$\therefore \Phi_{ab} = \sum_a U_{aa} \tilde{\Phi}_{a,b} \text{ and } \sum_a \Phi_{a,b} \overline{\Phi_{a,c}} =$$

$$\sum_{a,a_2} U_{aa} \tilde{\Phi}_{a,b} \overline{U_{aa_2} \tilde{\Phi}_{a_2,c}} = \sum_{a,a_2} U_{aa} U_{a_2a}^* \tilde{\Phi}_{a,b} \overline{\tilde{\Phi}_{a_2,c}}$$

$$= \sum_{a,a_2} (U^* U) \tilde{\Phi}_{a,b} \overline{\tilde{\Phi}_{a_2,c}} = \sum_a \tilde{\Phi}_{a,b} \overline{\tilde{\Phi}_{a,c}}$$

($\because U$ is a unitary matrix)

$$\text{Therefore, } \rho^B(\rho) = \rho^B(\tilde{\rho}).$$

That is, Bob's subsystem is always in the mixed state regardless of which observable Alice measures.

Our result: There is no BELL telephone!!

4.6.2 Perfect quantum copier.

X. A quantum copier does not exist.

Def 4.10 Given (i) an arbitrary state (the 'original' $|\psi\rangle \in H$) to be copied.

(ii) a state (the 'white page') $|w\rangle \in H$ to emerge as a copy.

A quantum copier is a linear transformation satisfying
 $K: |\psi\rangle \otimes |w\rangle \longmapsto |\psi\rangle \otimes |\psi\rangle$, for any $|\psi\rangle \in H$
 and a given fixed $|w\rangle \in H$.

Prop 4.11 A quantum copier cannot exist.

pt) $H = qH$ and $|0\rangle, |1\rangle$ and $\frac{1}{\sqrt{2}}(|1\rangle + |0\rangle)$.

obvne that

$$K(|0\rangle \otimes |w\rangle) = |0\rangle \otimes |0\rangle$$

$$K(|1\rangle \otimes |w\rangle) = |1\rangle \otimes |1\rangle$$

$$K\left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes |w\rangle\right) = \frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes \frac{|1\rangle + |0\rangle}{\sqrt{2}},$$

since K is linear.

$$K\left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes |w\rangle\right) =$$

$$K\left(\frac{1}{\sqrt{2}}(|1\rangle \otimes |w\rangle) + \frac{1}{\sqrt{2}}(|0\rangle \otimes |w\rangle)\right) =$$

$$\frac{1}{\sqrt{2}}(K(|1\rangle \otimes |w\rangle) + K(|0\rangle \otimes |w\rangle)) =$$

$$\frac{1}{\sqrt{2}}(|1\rangle \otimes |1\rangle + |0\rangle \otimes |0\rangle) \neq$$

$$\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes \frac{|1\rangle + |0\rangle}{\sqrt{2}} \quad \square$$

Remark Note that there can be devices that copy particular states as specified in Def 4.10
 $\Rightarrow$ refer to chapter 5.

In summary.

- ① BELL inequality
- ② CHSH inequality
- ③ There is no BELL telephone
- ④ A quantum-copier cannot exist. 